

Twelve Years of Cyber Resilience: Analyzing Cyberattacks in Saudi Arabia (2012-2024)

Fatemah Alharbi ¹

¹ Computer Science Department, Taibah University, Yanbu, Al-Madinah Province 46522, Saudi Arabia

Abstract – Over the last decade, Saudi Arabia has become a focal point in the global cybersecurity landscape, facing a rising number of cyberattacks linked to its rapid digitalization, particularly under the Vision 2030 initiative. This paper explores the evolution of cyber warfare and cybercrime targeting Saudi Arabia, documenting major cyberattacks from 2012 to 2024. It evaluates the impacts of these breaches, ranging from financial losses to threats to national security, and highlights the broader implications for the country's economy and global standing. The analysis extends to the government's responses, including the establishment of the National Cybersecurity Authority (NCA), implementation of advanced cyber laws, and strategic partnerships aimed at strengthening cyber defences. Saudi Arabia's improved standing in the Global Cybersecurity Index underscores its dedication to addressing vulnerabilities and enhancing digital security. However, the paper emphasizes that ongoing challenges remain, particularly as the kingdom navigates a rapidly evolving threat landscape. The study not only assesses Saudi Arabia's cybersecurity progress but also offers insights into global cyber defence strategies, underscoring the importance of continued vigilance, adaptive policies, and international cooperation to safeguard digital infrastructure in the future.

Keywords – Cyberattacks, Saudi Arabia, cybersecurity trends, cyber resilience.

DOI: 10.18421/TEM142-77

<https://doi.org/10.18421/TEM142-77>

Corresponding author: Fatemah Alharbi,
Computer Science Department, Taibah University, Yanbu,
Al-Madinah Province 46522, Saudi Arabia

Email: fmhharbi@taibahu.edu.sa

Received: 11 August 2024.

Revised: 04 March 2025.

Accepted: 27 March 2025.

Published: 27 May 2025.

 © 2025 Fatemah Alharbi; published by UIKTEN. This work is licensed under the Creative Commons Attribution-NonCommercial-NoDerivs 4.0 License.

The article is published with Open Access at <https://www.temjournal.com/>

1. Introduction

In the shadow of the burgeoning digital era, the Kingdom of Saudi Arabia stands at the forefront of a technological renaissance, propelled by its ambitious Vision 2030 [1]. This vision not only aims to diversify the economy beyond oil but also to integrate advanced digital technologies into the fabric of Saudi society. However, with great digitalization comes the heightened risk of cyber threats [2], [3], [4], which have emerged as a critical challenge to national security, economic stability [5], [6], and societal trust. Over the past decade, Saudi Arabia has become a focal point for cyberattacks, reflecting a global trend where cyber warfare and cybercrime have become instruments of power, disruption, and coercion.

The significance of cybersecurity in Saudi Arabia cannot be overstated. As one of the world's largest oil producers and a leading economic powerhouse in the Middle East, the kingdom holds strategic geopolitical and economic positions that make it an attractive target for cyber threat actors. The attack on Saudi Aramco in 2012 [3], one of the most devastating cyberattacks in history, served as a stark wake-up call to the nation and the world about the vulnerabilities inherent in digital infrastructures and the potential ramifications of cyber warfare. This incident underscored the urgent need for robust cybersecurity measures and catalyzed a national and organizational shift towards strengthening digital defenses.

In the years following, Saudi Arabia has witnessed a dynamic evolution in the nature and frequency of cyberattacks. From state-sponsored espionage [7], [8] to financially motivated cybercrime [9], [10], the kingdom has faced a diverse array of cyber threats aimed at destabilizing its critical infrastructure, pilfering sensitive information, and undermining its economic and social fabric. The transition to remote work during the COVID-19 pandemic further exacerbated these vulnerabilities [11], leading to an unprecedented surge in cyber incidents. This evolving threat landscape necessitated an equally dynamic response from the Saudi government, which has since undertaken significant strides to bolster its cyber defense capabilities.

The establishment of the National Cybersecurity Authority (NCA) [12], the Saudi National Cybersecurity Authority's Computer Emergency Response Team (CERT) [13], and the introduction of stringent cyber laws underscore Saudi Arabia's commitment to creating a robust digital infrastructure. These efforts have significantly contributed to the nation's improved ranking (Second Globally) in the Global Cybersecurity Index [14], reflecting its strategic move towards ensuring a secure and resilient digital environment.

As this paper explores the intricate history of cyberattacks against Saudi Arabia over the last 12 years (2012-2024), it seeks to answer pivotal questions about the nature of digital threats, the evolution of cybersecurity strategies, and the implications of cyber warfare in a connected world. Through a meticulous analysis of past cyber incidents and governmental responses, this study aims to provide insights into the challenges and achievements of Saudi Arabia's cybersecurity initiatives, offering a lens through which the future of digital security in the kingdom and beyond can be seen. This exploration is not only a chronicle of past events but a forward-looking perspective on the ongoing battle in the cyber domain, where innovation, resilience, and international cooperation serve as the bulwarks against an ever-evolving threat landscape.

The remainder of the paper is organized as follows. The methodology and scope are explained in Section 2. The chronological progression of cyberattacks in Saudi Arabia is presented in Section 3. The challenges and future work are discussed in Section 4. The closely related work is stated in Section 5. The paper is concluded in Section 6.

2. Methodology and Scope

The methodology for this research paper is anchored in a theoretical literature review focused on cyberattacks in Saudi Arabia from the period of 2012 to 2024. The approach involved an extensive examination of available literature, including academic journals, industry reports, cybersecurity incident databases, and news articles. This comprehensive review aimed to synthesize a wide range of sources to construct a detailed narrative of the evolving cyber threat landscape in Saudi Arabia. Additionally, a comparative analysis was conducted to identify recurring attack patterns, assess the effectiveness of implemented defense mechanisms, and highlight emerging cybersecurity challenges.

2.1. Research Questions

The study focuses on providing answers to three crucial research questions as shown in Table 1. Addressing these research questions required a comprehensive analysis of cyberattacks in Saudi Arabia over the past twelve years. This involved an extensive literature review of cyber incidents, their impact on critical infrastructure, and the effectiveness of implemented security measures. By systematically investigating these areas, the study aims to provide a well-rounded perspective on the cybersecurity landscape of Saudi Arabia, highlighting key trends, emerging threats, and defensive strategies. This analysis serves as a valuable resource for policymakers, cybersecurity professionals, and organizations aiming to strengthen their cybersecurity posture against increasingly sophisticated cyber threats.

Table 1. Research questions

No.	Research Question	Motivation
RQ1	What is the trend of cyberattacks in Saudi Arabia over the period 2012-2024?	Tracking cyberattacks over time is crucial for understanding the evolving landscape of cybersecurity in Saudi Arabia. This longitudinal analysis is not only crucial for detecting patterns and trends in cyberattack methodologies but also in evaluating the effectiveness of the defensive measures implemented. By examining these trends over time, stakeholders can better strategize and allocate resources to fortify the nation's digital infrastructure against evolving cyber threats. This approach ensures a proactive rather than reactive cybersecurity framework, adapting to new challenges as they arise.
RQ2	What are the cybersecurity challenges that Saudi Arabia is facing?	Understanding the cybersecurity challenges that Saudi Arabia faces is crucial for developing effective strategies to protect against potential threats. The motive behind studying these challenges is to enhance national security, safeguard critical infrastructure, and ensure the economic and political stability of the country. By identifying and addressing these vulnerabilities, Saudi Arabia can better defend against the growing sophistication of cyberattacks and mitigate risks that could impact its governmental, economic, and social systems.
RQ3	What strategies have Saudi organizations and authorities implemented to defend against cyberattacks? Were they effective?	Understanding the strategies and regulations implemented by Saudi organizations and authorities helps in evaluating their cybersecurity resilience and preparedness. The motive is to assess the effectiveness of these strategies in mitigating risks, ensuring that sensitive information remains secure, and that infrastructure is protected against potential cyber threats. This knowledge is vital for continuous improvement and adaptation of security protocols to counter evolving cyber threats effectively.

2.2. Research Model

The methodology for the systematic literature review in this research is based on a model by [15] and incorporates key principles for conducting the study effectively as outlined by [16].

Recently, the focus has been on gathering information on cyberattacks from various literature and articles, emphasizing web security. Eventually, the scope in this paper is narrowed to specific cybersecurity domains, aligning with the findings of [17] and [18], over the last twelve years in Saudi Arabia. The rigorous inclusion and exclusion criteria used for this review are detailed in Table 2.

Several key steps were taken in the process of collecting articles and literature, utilizing major databases such as Elsevier, Journal of Science, ResearchGate, ScienceDirect, ACM Digital Library, IEEEExplore, SpringerLink, and various Internet sources. The search terms fell into specific categories such as cybersecurity, cybercrime, and cyberattack trends, challenges, and strategies in Saudi Arabia, using both individual and combined terms through Boolean operators "AND" and "OR." The strategy also included testing synonyms to ensure comprehensive coverage of recent and relevant articles and publications on the subject.

Table 2 displays the systematic literature and information search approach.

Table 2. Considerations for inclusion and exclusion in this review

No.	Inclusion Criteria	Exclusion Criteria
1	Articles are in the field of cybersecurity	Articles outside the cybersecurity field
2	Articles are solely written in Arabic and English languages	Articles written in languages other than Arabic and English
3	Cyberattack year was limited to the 2012 to 2024 range	Cyberattack year before 2012
4	Articles that are peer-reviewed, proceedings from conferences, and technical reports	Television commentaries, blogs, and newspaper reports
5	Keywords such as cyberattack, cybercrime, or cybersecurity must be included in the title of the article	Articles with titles that do not include any of the three keywords
6	Reports aligned with statements from individuals, organizations, or nation	Cyberattacks with no formal statement from individuals, organizations, or nation
7	Earlier reviews in the field that have explored the strengths and weaknesses of cybersecurity measures	Editorials focusing on the necessity for research within the field often debate the arguments surrounding the need for further investigation and development

Figure 1 shows the Systematic literature review methodology. Initially, a total of 112 articles were identified through database and Internet searches based on their titles.

Upon reviewing their abstracts, 17 articles were excluded. Following a full text screening, 87 articles were determined to be suitable for inclusion in the study.

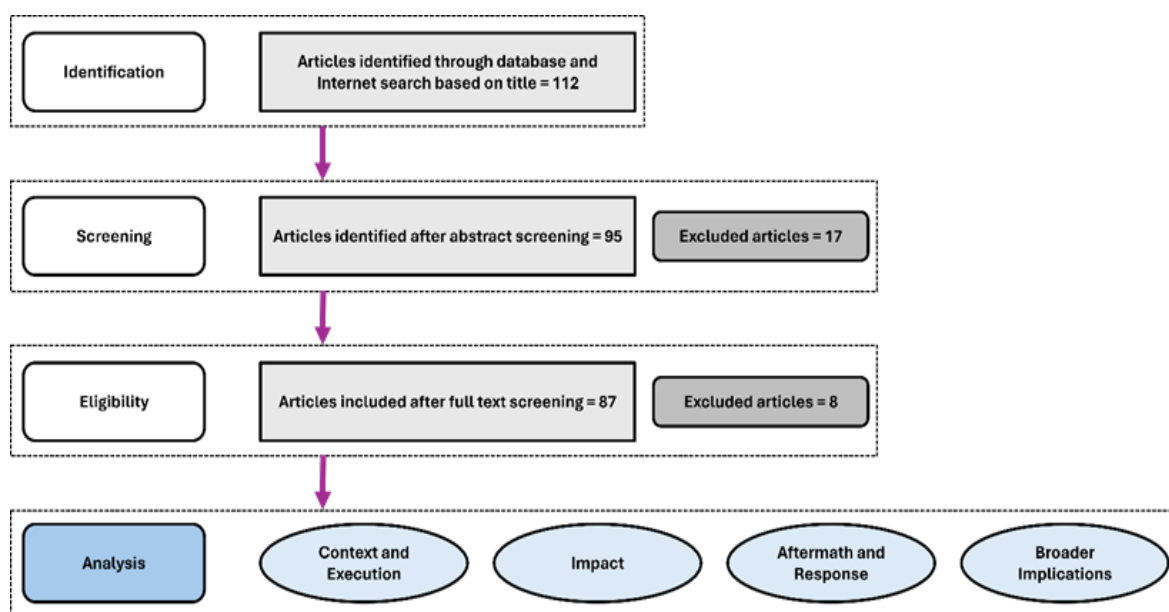


Figure 1. Systematic literature review methodology

2.3. Metrics

The analysis was structured around four critical metrics (as shown in Figure 1) to ensure a holistic examination of each cyber incident:

- 1) **Context and Execution:** This metric delves into the background of each cyberattack, exploring the tactics, techniques, and procedures employed by attackers. It seeks to understand the operational aspects of how attacks were carried out and the technological vulnerabilities exploited.
- 2) **Impact:** This dimension assesses the immediate and long-term consequences of cyberattacks on the targeted entities and broader sectors. It considers both tangible impacts, such as financial losses and operational disruptions, and intangible effects, including reputational damage and psychological impacts on stakeholders.
- 3) **Aftermath and Response:** This metric examines the responses by targeted organizations and the national cybersecurity framework to mitigate the impact of attacks and prevent future incidents. It includes an analysis of recovery efforts, changes in cybersecurity policies, and the deployment of new defense technologies.
- 4) **Broader Implications:** Beyond the immediate context of each attack, this dimension explores the wider implications for national security, economic stability, and international relations. It considers how incidents have shaped cybersecurity strategies, legislation, and international cooperation against cyber threats.

Through this methodological approach, the paper intends to contribute to the broader discourse on cybersecurity, offering valuable lessons learned and best practices that can inform future defense strategies not only for Saudi Arabia but also for the global community facing similar threats.

The scope of this paper encompasses a detailed analysis of significant cyberattacks targeting Saudi Arabia from 2012 to 2024. The period is chosen to capture a critical phase in the kingdom's cybersecurity evolution, marked by high-profile incidents like the Shamoon virus attacks and the increasing sophistication of cyber threats. While the focus is on attacks within Saudi Arabia, the analysis is situated within a global context to understand the international dimensions of cybersecurity challenges and solutions. The paper aims to provide insights into the strategic, operational, and policy responses to cyber threats, contributing to the broader discourse on cybersecurity resilience and governance.

3. The Landscape of Cyber Threats in Saudi Arabia (2012-2024)

The cyber threat landscape in Saudi Arabia from 2012 to 2024 presents a complex and evolving narrative marked by a series of high-profile cyberattacks, shifts in threat actor tactics, and the growing sophistication of cybersecurity measures. This period reflects a critical phase in the kingdom's digital transformation and its implications for national security, economic policies, and global cyber diplomacy.

3.1. Major Cyberattacks: A Chronology

In this section, the study highlights examples of incidents that underscore the multifaceted nature of cyber threats facing Saudi Arabia, spanning from destructive malware and espionage to sophisticated phishing and ransomware attacks. Each incident has prompted enhancements in cybersecurity measures, legislative actions, and the adoption of advanced technologies to protect against future threats. The chronological progression of these attacks (Table 3) highlights the evolving challenge of cybersecurity and the perpetual arms race between cyber defenders and threat actors.

Table 3. Timeline of major cyberattacks in Saudi Arabia (2012-2024)

Year	Type	Name	Main Target	References
2012	Malware	Shamoon	Saudi Aramco	[3], [19], [20]
2014	Campaigns	Espionage/Spear-Phishing Campaigns	Saudi Aramco	[21], [22], [23]
2017	Malware	Shamoon 2	Government agencies and Critical infrastructures	[24], [25], [26], [27]
2019	Attack	Drone Strikes	Saudi Aramco	[28], [29]
2020	Attack	Ransomware	All sectors	[30], [31]
2121	Remote Work Vulnerabilities	-	All sectors	[32], [33]
2022	Campaigns	Espionage/Spear-Phishing	Government agencies	[34], [35], [36]
2023	Campaigns	-	Critical infrastructures	[37], [38]
2024	Advances Persistent Threat (APT)	-	Government agencies and Energy sector	[39]

3.1.1. (2012) Shamoon Virus Attack on Saudi Aramco

The Shamoon virus attack on Saudi Aramco, the world's largest oil company at the time, in August 2012 is one of the most notable cyber incidents in the history of cybersecurity [3], [19], [20], particularly highlighting the vulnerabilities of critical infrastructure to cyber warfare.

This sophisticated cyberattack was not just a wake-up call for Saudi Arabia but for the world, highlighting how cyberattacks could cause physical disruptions to critical infrastructure. Shamoon's ability to spread rapidly within the network and overwrite files with garbage data, rendering machines inoperable, showcased a new level of cyber warfare.

Context and Execution: The attack was meticulously planned and executed, leveraging the Shamoon malware, a piece of destructive computer virus designed to spread across the network of the oil giant. The malware (as explained in [40] and [41]) operated by infecting workstations and servers, with the primary intention of wiping data and rendering the machines inoperable. It achieved this by overwriting the master boot record (MBR) of infected computers, making it impossible for them to boot up and effectively turning them into expensive paperweights. One of the distinctive features of the Shamoon virus was its capability to report back the details of the wiped data to the attackers before rendering the machines useless. This feature indicates that beyond mere destruction, there was also an intent to intimidate and perhaps gather intelligence.

Impact: The immediate impact of the attack was staggering: approximately 35,000 computers were incapacitated, leading to significant data loss and a massive operational disruption. For Saudi Aramco, this meant a temporary halt in operations and a monumental effort in damage control and recovery. The company had to isolate its entire network to contain the spread of the virus, and the recovery process involved not only the cleaning and restoration of infected systems but also a strategic overhaul of its cybersecurity posture.

Aftermath and Response: In the aftermath of the Shamoon attack, Saudi Aramco undertook an extensive review of its cybersecurity defenses [42]. The company accelerated the implementation of advanced threat detection systems, enhanced its cyber incident response capabilities, and bolstered its IT infrastructure to defend against future attacks. This incident also served as a catalyst for the Saudi government to prioritize national cybersecurity, leading to the establishment of dedicated cybersecurity organizations [12], [13] and initiatives aimed at protecting critical infrastructure and ensuring the resilience of the kingdom's digital landscape.

Broader Implications: The Shamoon attack underscored the evolving nature of cyber threats and the need for continuous vigilance and adaptation in cybersecurity strategies. It highlighted the potential of cyberattacks to cause physical damage and disrupt critical services, a reality that has since guided the development of cybersecurity policies and practices not only in Saudi Arabia but globally. Moreover, the attack has been a case study in cybersecurity circles [34], [44], illustrating the importance of not only defensive measures but also the need for robust incident response plans and the capability to recover from such incidents. The Shamoon virus attack remains a pivotal moment in the history of cyber warfare, serving as a stark reminder of the vulnerabilities inherent in digital infrastructures and the continuous need for improvement in cyber defenses.

3.1.2. (2014) Aramco Spear-Phishing Campaigns

The 2014 Aramco Spear-Phishing Campaigns [21], [22], [23] represent a significant evolution in the cyber threat landscape faced by Saudi Aramco, moving from broad, destructive attacks to more targeted and sophisticated espionage efforts. After the widespread damage inflicted by the Shamoon virus in 2012, attackers began employing spear-phishing as a method to infiltrate the company's defenses, indicating a strategic shift towards stealthier, information-gathering operations.

Context and Execution: Spear-phishing campaigns involve sending fraudulent emails that appear to come from a trusted source but contain malicious attachments or links. The goal is to deceive the recipient into disclosing sensitive information or installing malware. In the case of Aramco, these campaigns were meticulously crafted to target specific employees, likely with access to valuable corporate information or operational systems. This method demonstrates a profound understanding of human psychology and organizational structures, leveraging social engineering to bypass technical security measures. Specifically, the attackers utilized emails that mimicked internal communications and messages from known business contacts, embedding malicious software within seemingly benign documents or directing employees to compromised websites. Once an employee interacted with these deceitful elements, attackers could gain unauthorized access to Aramco's network, allowing them to conduct surveillance, exfiltrate data, or establish persistent access for future operations.

Impact: Unlike the overt disruption caused by Shamoon, the spear-phishing campaigns aimed at stealth and persistence, avoiding immediate detection to maximize intelligence gathering. The impact of these campaigns extended beyond the immediate security breaches, undermining trust within the organization and necessitating a comprehensive review of email security practices, employee training, and incident response protocols. According to security report by McAfee [45], Saudi Arabia incurred costs of approximately 4 billion riyals due to cyberattacks in that year.

Aftermath and Response: In response to these incidents, Saudi Aramco and other entities within the kingdom accelerated the implementation of advanced cybersecurity measures [42]. These included enhanced email filtering technologies, the adoption of secure email gateways, and the deployment of anti-phishing training programs designed to educate employees about the dangers of spear-phishing and the importance of vigilance when handling electronic communications. In 2014, the establishment of the National Centre for Cybersecurity Technology (now known as the National Cybersecurity Authority (NCA) [12]) was realized, serving as a research and development body focused on tackling issues related to network, software, and information security. Furthermore, these attacks underscored the necessity for robust endpoint detection and response (EDR) systems [46] capable of identifying and mitigating threats that bypass initial security perimeters. The integration of behavioral analytics and machine learning algorithms became critical in detecting anomalous activities indicative of a breach, enabling faster response times and minimizing potential damage.

Broader Implications: The attacks highlight a critical shift in the cyber threat landscape, where targeted, sophisticated attacks pose a significant risk to corporate and national security. These incidents catalyzed a broader recognition of the importance of cybersecurity hygiene, employee awareness, and the need for continuous investment in advanced defensive technologies. As cyber threats continue to evolve, the lessons learned from these spear-phishing campaigns remain relevant, emphasizing the need for a multi-layered security approach that encompasses both technical defenses and human factors. The ongoing battle against cyber threats requires not only technological innovation but also a culture of security awareness that empowers individuals to act as a first line of defense.

3.1.3. (2017) Shamoon 2

The 2017 reemergence of the Shamoon virus, often referred to as Shamoon 2 [24], [25], [43], [47], [48] underscores a critical escalation in the cyber threat landscape targeting Saudi Arabia.

This iteration of the virus demonstrated a nuanced understanding of cybersecurity defenses and exploited them with refined tactics, significantly impacting the kingdom's critical infrastructure and government entities.

Context and Execution: Shamoon 2 displayed enhanced capabilities compared to its predecessor, incorporating features designed to increase its destructiveness and evade detection. This version was capable of spreading more efficiently across networks, leveraging stolen credentials to move laterally and targeting specific data types for destruction. Its deployment in waves indicated a calculated approach to maximize disruption, suggesting that the attackers had specific knowledge of their targets' operational cycles and possibly inside information about their network architectures.

Impact: The impact of Shamoon 2 was profound, affecting not just the operational capabilities of the targeted entities but also signaling a stark warning to other nations and industries about the evolving nature of cyber threats [49]. The attack disrupted government services and operations, causing data loss and significant downtime [50]. Moreover, it served as a psychological tool, instilling fear and uncertainty about the security of critical infrastructure and the effectiveness of existing cybersecurity measures. The incident marked the most severe in a series of cyberattacks targeting petrochemical facilities in Saudi Arabia. In January 2017, the National Industrialization Company (known as Tasnee) and one of the few privately owned Saudi petrochemical firms, experienced a significant shutdown of its computer systems [26]. Simultaneously, similar disruptions occurred at Sadara Chemical Company, situated 15 miles away, which is a collaboration between Saudi Aramco and Dow Chemical [27].

Aftermath and Response: The aftermath of Shamoon 2 prompted a swift and comprehensive response from both the affected entities and the Saudi Arabian government. This involved an immediate assessment of the damage, the implementation of emergency recovery plans, and a thorough investigation into the breach mechanisms. In response to the attack, there was a marked increase in the adoption of advanced cybersecurity technologies, including behavior analytics, Artificial Intelligence (AI) for threat detection, and enhanced incident response capabilities. Additionally, there was a renewed focus on cybersecurity training for employees, emphasizing the importance of vigilance and adherence to security protocols. More importantly, the Saudi NCA [51] was established by royal decree on 31 October 2017 by King Salman bin Abdulaziz. Since its establishment, the NCA has been proactive in enhancing the nation's cybersecurity framework to protect critical infrastructure and key sectors in alignment with Vision 2030 [1].

Broader Implications: The reemergence of Shamoon served as a catalyst for broader discussions on national and international cybersecurity strategies. It highlighted the necessity for ongoing investment in cybersecurity research and development, the importance of international cooperation in combating cyber threats, and the need for a proactive and dynamic approach to cybersecurity. The attack also influenced regulatory frameworks, leading to the development and enforcement of stricter cybersecurity standards and guidelines for critical infrastructure protection, e.g., Request For Comments (RFCs) by the National Institute of Standards and Technology (NIST) [52], ISO/IEC 27001:2022 by the International Organization for Standardization (ISO) [53], and the Domain Name System (DNS) standards by the Internet Assigned Numbers Authority (IANA) [54]).

3.1.4. (2019) Oil Facility Drone Strikes and Cyber Component

In conjunction with physical drone attacks on oil facilities in 2019, there were reports of cyberattacks aimed at exacerbating the damage and disruption. These coordinated attacks signaled a new era of hybrid warfare, where cyber operations are integrated with physical attacks to amplify their impact.

Context and Execution. In September 2019 [28], [29], [55], a series of drone strikes targeted key oil facilities in Saudi Arabia, leading to substantial disruptions in global oil supply. What made these attacks particularly noteworthy was the reported integration of a cyber component [56], aimed at amplifying the physical damage and complicating the recovery processes. This blend of physical and cyber tactics indicates a sophisticated understanding of the target's vulnerabilities and represents a shift towards hybrid warfare strategies, where cyber operations are used to enhance the effectiveness of physical attacks.

Impact. The immediate impact of the drone strikes was a temporary halving of Saudi oil production, a significant blow to the global energy market. The accompanying cyber operations, though less publicized, were aimed at hindering the response and recovery efforts, showcasing the potential for cyberattacks to exacerbate the effects of physical assaults on critical infrastructure. This incident had economic ramifications --- Saudi Arabia experienced a significant reduction in its oil output, decreasing from 9.8 million barrels to approximately 4.1 million barrels per day. This decline resulted in a shortfall of 5.7 million barrels daily, equivalent to about 5% of the worldwide oil production [57], [58]). This attack also raised serious questions about the security of critical energy infrastructure and the need for comprehensive protective measures that address both physical and cyber threats.

Aftermath and Response: The response to the attacks, particularly the cyber component, involved a rapid mobilization of national and international cybersecurity resources to assess the damage, mitigate any ongoing threats, and restore operations [59]. It underscored the importance of resilience in critical infrastructure, highlighting the need for robust cybersecurity defenses [60], quick incident response capabilities, and the ability to recover from sophisticated hybrid attacks.

Broader Implications. The 2019 oil facility attacks marked a pivotal moment in the understanding of modern warfare and critical infrastructure protection. It demonstrated that adversaries are increasingly likely to employ hybrid tactics that combine physical and cyber attacks [61], [62] to achieve strategic objectives. This realization has prompted a reevaluation of security strategies across critical sectors, with an emphasis on the integration of physical security measures and cybersecurity practices. For instance, the North Atlantic Treaty Organization (known as NATO) has developed strategies to address its involvement in combating hybrid warfare, ensuring readiness to protect the alliance and every member against any form of threat [63], [64]. Similarly, the Russian Ministry of Defense has articulated a strategic approach to shield the nation from hybrid threats [65]. The incident has also led to increased calls for international cooperation to address the threats posed by such hybrid attacks, recognizing that the security of global critical infrastructure is a shared concern that requires collective action. The evolution of threats highlighted by the 2019 attacks continues to influence how nations, corporations, and international bodies conceptualize and implement security in an increasingly interconnected and technologically advanced world.

3.1.5. (2020) Increase in Ransomware Attacks

The increase in ransomware attacks in 2020 marked a significant escalation in cyber threats globally, including within Saudi Arabia [30], [31]. This period saw a notable rise in such attacks, driven in part by the vulnerabilities introduced by the widespread shift to remote work due to the COVID-19 pandemic.

Context and Execution: The proliferation of ransomware in 2020 exploited the rapid digital transformation and the increased reliance on digital networks for remote work. Cybercriminals targeted not just large corporations but also small businesses, healthcare institutions, and educational facilities, leveraging the urgency and chaos of the pandemic environment. The attacks involved encrypting the victim's data and demanding ransom for the decryption key, often accompanied by threats of public data release.

Impact: As the coronavirus pandemic intensified, Saudi Arabia experienced a significant surge in cybersecurity activities [66]. The kingdom observed a 4% rise in malware incidents and a significant surge of 378 in ransomware occurrences [30]. The impact of these ransomware attacks was multifaceted. Victims faced substantial financial burdens with total costs of \$5.97 million U.S. dollars [31], not only from the ransoms paid but also from the downtime and loss of productivity. Additionally, the threat of data breaches added a layer of complexity, risking personal privacy and corporate intellectual property. The attacks also caused operational disruption. Critical services, especially in healthcare and education, were severely disrupted, affecting societal functions at a critical time.

Aftermath and Response: The response to the surge in ransomware attacks led to significant adaptations in cybersecurity practices. There was a heightened focus on basic cybersecurity practices, such as regular backups, patch management, and phishing awareness training. The NCA published a number of controls, frameworks, and guidelines at the national level focused on cybersecurity [67], [68]. These measures aim to strengthen the cybersecurity landscape within organizations. Organizations increasingly adopted advanced security solutions, including endpoint detection and response (EDR) systems (such as [69], [70]), to identify and mitigate threats proactively. Interestingly, the rise in attacks fostered greater collaboration between organizations and national cybersecurity authorities, sharing threat intelligence and best practices to counteract the ransomware epidemic [71].

Broader Implications: The 2020 increase in ransomware attacks highlighted the evolving cyber threat landscape and the need for continuous vigilance and adaptation in cybersecurity strategies. It underscored the importance of resilience planning, not just in IT infrastructure but across all organizational functions, to withstand and recover from cyberattacks. This period served as a catalyst for strengthening cybersecurity frameworks and policies, emphasizing the critical role of cybersecurity in ensuring operational continuity and trust in digital systems.

3.1.6. (2021) Remote Work Vulnerabilities

The vulnerabilities introduced by the shift to remote work in 2021, amidst the ongoing COVID-19 pandemic, marked a significant change in the cybersecurity landscape.

In Saudi Arabia, 84% of those who experienced a cyberattack reported an increase in incidents, attributing the rise to the higher number of individuals working remotely [32].

As organizations and employees adapted to new ways of working outside traditional office environments, cybercriminals quickly took advantage of emerging security gaps. This period necessitated a reevaluation of cybersecurity practices to address the unique challenges posed by remote work setups.

Context and Execution: The rapid transition to remote work in Saudi Arabia forced many organizations to rely heavily on virtual private networks (VPNs), with more than 12.7 million downloads [72], and cloud services - the market for cloud business process services (BPaaS) had a value of \$495 million U.S. dollars in the Middle East and North Africa (MENA region) [73]. These technologies, while facilitating remote access, also introduced new vulnerabilities [74], [75], particularly when not configured or secured properly. Cyber attackers exploited these weaknesses, launching attacks aimed at stealing credentials, intercepting data, and infiltrating corporate networks. Furthermore, the shift also highlighted the human factor as a critical vulnerability. According to IBM [76], human error is #1 cybersecurity threat to businesses in 2021. Employees working from home were more likely to use personal, less secure devices and networks for work-related activities. Phishing and social engineering attacks increased, with attackers exploiting the uncertainty and fear surrounding the pandemic to trick individuals into revealing sensitive information or downloading malware. In addition, many organizations found their existing cybersecurity policies and training inadequate for the realities of remote work. The lack of clear guidelines on secure remote work practices and the absence of regular security awareness training for employees exacerbated the risks, leaving staff unprepared to recognize or respond to cybersecurity threats effectively.

Impact: There has been a noticeable rise in phishing campaigns targeting individuals and organizations within the kingdom. According to VMware [32], phishing attacks and outdated security technologies were identified as primary factors, contributing to 15.5% of security breaches. These attacks often serve as the entry point for more severe cyber incidents, including data breaches and ransomware attacks. By deceiving employees into divulging login credentials or downloading malware, attackers gain unauthorized access to sensitive systems and information.

Also, social engineering attacks have leveraged insider knowledge and trust to bypass security measures. Beyond direct cyber incidents, the human factor contributes to broader financial and economic consequences. According to Positive Technologies [33], attacks targeting individuals (affecting 96% of people and 41% of organizations) are highly prevalent, with human error contributing to over 80% of cyber incidents. This trend, highlighted in Verizon's yearly analysis [77], is notably evident in Saudi Arabia, underscoring the critical role of human factors in cybersecurity breaches.

Aftermath and Response: In response to these vulnerabilities, organizations began implementing more robust cybersecurity measures tailored to the remote work environment. This included the adoption of multi-factor authentication (MFA) to secure access to corporate resources, the deployment of endpoint protection solutions to secure remote devices (e.g., a high-standard and secure password policy), the enhancement of network security through advanced threat detection and response capabilities, and prohibiting remote access from outside Saudi Arabia [78]. Moreover, there was a significant emphasis on cybersecurity awareness training programs and institutes such as Taqnia Institute [79], CyberIC program by NCA [80], and the cybersecurity initiatives by Saudi CERT [13]. These programs specifically designed to address the risks associated with remote work. Organizations also revised their cybersecurity policies to include comprehensive guidelines on secure remote work practices, data protection, and incident reporting adhering to the NCA Essential Cybersecurity Controls (ECCs) [81], the Operational Technology Cybersecurity Controls (OTCCs) [82], and the Critical Systems Cybersecurity Controls (CSCCs) [78].

Broader Implications: The experience of 2021 underscored the need for a flexible and dynamic approach to cybersecurity, one that can adapt to changes in how and where work is conducted. It highlighted the importance of cybersecurity resilience, not just in terms of technology but also in fostering a culture of security awareness among employees. As remote work continues to be a significant aspect of the modern workplace [83], the lessons learned during this period will undoubtedly shape future cybersecurity strategies, emphasizing the need for ongoing vigilance, adaptation, and education in the face of evolving cyber threats.

3.1.7. (2022) Cyber Espionage Campaigns

The cyber espionage campaigns in 2022 marked a sophisticated evolution in the cyber threat landscape, particularly impacting Saudi Arabia [33], [34], [35], [36].

These campaigns were characterized by their complexity, targeting government agencies, critical infrastructure, and strategic sectors to gather intelligence, disrupt operations, or influence geopolitical dynamics.

Context and Execution: Cyber espionage activities in 2022 utilized a range of advanced tactics, techniques, and procedures. These attacks included malware blocked in industrial control systems (ICS) [35], the deployment of Advanced Persistent Threats (APTs) that remain undetected within networks for extended periods, spear-phishing attacks aimed at high-value targets, and the exploitation of zero-day vulnerabilities [33]. The same report by Positive Technologies classifies the attacking groups into three categories: (1) APT groups; (2) Hacktivists; and (3) Ransomware attackers --- APT groups conducted 40% of successful cyberattacks. The objective was often to exfiltrate sensitive data, monitor communications, and gain a strategic advantage over Saudi Arabia in terms of intelligence and geopolitical leverage.

Impact: In Middle Eastern countries including Saudi Arabia, government bodies emerge as prime targets for cybercriminals, making up 22% of all organizational cyberattacks. These incidents primarily lead to two major repercussions: The interruption of essential operations, affecting 36% of cases, and the unauthorized disclosure of sensitive data, occurring in 28% of instances. Furthermore, the attacks targeted industrial sector organizations using mostly social engineering channels (33%), remote administration tools (known as RATs) (62%), and wipers (31%). The "Hi-Tech Crime Trends 2022/2023" report by Group-IB [36] indicates that within the Persian Gulf area, Saudi Arabia faced the second highest percentage of cyberattacks, with 29%, marking it as a significant target in the region. Regarding financial losses, the Middle East (a cluster sample of companies located in Saudi Arabia and the United Arab Emirates (UAE)) ranks as the region with the second-highest average cost resulting from data breaches worldwide, reaching approximately \$7.46 million U.S. dollars in 2022 [84].

Aftermath and Response: In response to the heightened threat of cyber espionage, Saudi Arabia bolstered its cybersecurity posture through several measures. These included enhancing the capabilities of national cybersecurity agencies, increasing collaboration with international partners for intelligence sharing, and investing in advanced threat detection and response technologies [67], [81], [82]. Additionally, efforts to raise awareness about cybersecurity best practices among government employees and critical sector workers have been made [85], [86], as awareness is considered a critical metric in mitigating the risk of future breaches.

Broader Implications: The cyber espionage campaigns of 2022 underscored the increasingly sophisticated and stealthy nature of cyber threats faced by nations like Saudi Arabia. These incidents highlighted the necessity for continuous vigilance, advanced cybersecurity capabilities, and a proactive approach to national and organizational cybersecurity. Moreover, they illustrated the importance of cybersecurity as an integral component of national security and international diplomacy.

3.1.8. (2023) Critical Infrastructure Cyberattacks

The cyberattacks on critical infrastructure in 2023 represent a concerning evolution in the scale and sophistication of cyber threats, with far-reaching implications for national security, public safety, and economic stability. These incidents highlight the strategic targeting of systems and networks essential for the functioning of society, including energy grids, water supply systems, and transportation networks.

Context and Execution: The 2023 cyberattacks on critical infrastructure were characterized by their precision and sophistication, indicating a high level of planning and knowledge of the targeted systems. Attackers deployed advanced techniques such as ransomware to lock out operators, malware designed to damage physical equipment, and stealthy infiltrations aimed at espionage and long-term access [87]. These tactics suggest the involvement of highly skilled actors, potentially with state sponsorship, aiming not just for immediate disruption but for strategic advantage. In 2023, 83% of successful cyberattacks were specifically targeted rather than opportunistic. Government agencies were prime targets for cybercriminals, with these institutions comprising 22% of the total attacks directed at various organizations [33]. According to Sirar by STC [37], the average number of cyberattacks targeting Saudi Arabia has increased by 121%. The same report shows interesting statistics: four malware families—Emotet, Upatre, Qakbot, and Qbot—account for 68% of the cyberattacks. The most common tactics in these cyberattacks included the use of scripting and PowerShell scripts, accounting for 14%, followed by tools that utilized application layer protocols at 13%, and process discovery techniques at 12%.

Impact: The impact of these attacks was profound, affecting not just the direct operations of the targeted facilities but also the broader societal functions that depend on them. Technical reports show that the predominant motivation behind these cyber threats was financial [37]. Additionally, the number of Distributed Denial of Service (DDoS) attacks has risen by 19% compared to 2022.

Sirar by STC reported that 32% and 24% of reported attacks were of type brute force and malicious connection, respectively. The report shows that 12% of cyberattacks had an impact on the government and military sectors, the telecommunication sector, and financial services.

Aftermath and Response: In response to these threats, the government and private sector operators of critical infrastructure have intensified their focus on cybersecurity resilience. Measures implemented included:

- Enhanced monitoring and detection: Deploying advanced surveillance systems to detect and respond to anomalies in real-time.
- Robust incident response plans: Establishing and regularly updating incident response protocols to ensure swift action can be taken to mitigate the impact of an attack.
- Investment in cybersecurity talent: Building teams of cybersecurity experts dedicated to protecting critical infrastructure and training existing staff on best practices.
- Public-private partnerships: Strengthening collaboration between government agencies and private sector operators to share threat intelligence and best practices.

Broader Implications: The targeting of critical infrastructure highlights the broader implications of cyber threats on national security and underscores the necessity for a comprehensive and coordinated approach to cybersecurity. It also emphasizes the need for international cooperation in combating cyber threats, given their ability to transcend national borders and impact global supply chains and services.

3.1.9. (2024) Advanced Persistent Threats (APTs)

In 2024, Saudi Arabia has emerged as a key target for complex cyberattacks executed by criminal entities known as APT groups. These groups, such as APT15, APT33, and OilRig, strategically execute multi-phase attacks on specific sectors for religious [89], political, economic, and military reasons. Occasionally, these APT groups may also engage in acts of hacktivist or deliberate sabotage.

Context and Execution: The APT groups compiled detailed profiles of their intended targets, including the identities of employees, software specifications, and existing network weaknesses. Attackers conducted extensive network scanning (e.g., active scanning [90]) to identify suitable targets. In a similar vein, APT35 has focused its efforts on Saudi Arabia, specifically collecting employee phone numbers to facilitate mobile malware attacks.

The APT groups utilized these numbers to dispatch messages containing links to mobile malware aimed at espionage and data theft. They tracked the IP addresses [91] and geographical locations [92] of individuals visiting their phishing websites and pinpointed valuable email addresses to initiate their attacks. Previously, the Hexane group had also identified key personnel such as managers and employees within HR and IT departments of their targeted organizations for more focused attacks [39].

Impact: The profile of industries frequently targeted by APT groups in Saudi Arabia is quite predictable [88]. Mostly, APT groups have targeted government institutions at least once, and a significant majority, 69%, have specifically attacked the energy sector [39]. According to the Positive Technologies report, 56%, 25%, and 38% of the APT groups successfully installed malware on the targeted victims, injected web shells into compromised nodes, and extracted credentials (e.g., passwords) from specialized stores, respectively.

Aftermath and Response. In response to the APT group attacks, Saudi Arabia has taken comprehensive measures to bolster its cybersecurity defenses, particularly in critical sectors like petrochemicals. The aftermath of such cyberattacks typically involves several key response steps and strategic initiatives. The immediate focus is on containing the breach to prevent further damage. This involves disconnecting infected systems, blocking malicious IP addresses, and implementing additional security measures to isolate the attack vector. Furthermore, cybersecurity teams conduct detailed forensic investigations to understand the attack's scope, the methods used by the attackers, and the data or systems compromised. This step is crucial for identifying vulnerabilities and preventing future breaches. Also, efforts are made to restore affected systems and data from backups, ensuring minimal disruption to operations. This process is critical in maintaining business continuity and mitigating the economic impact of the attack.

Broader Implications: The response to APT attacks highlights the need for an ongoing commitment to cybersecurity as a top organizational priority. It emphasizes the dynamic nature of cyber threats and the need for continuous adaptation and vigilance in cybersecurity practices. The measures taken post-APT attacks not only aim to recover from the immediate damage but also to prepare better for future threats, ensuring resilience and security in a digitally connected world. These responses and strategies help shape a more robust national cybersecurity posture, crucial for safeguarding critical economic and infrastructure assets.

4. Challenges and Future Directions

The realm of cybersecurity, particularly in response to evolving threats such as APTs, presents a series of ongoing challenges and delineates clear pathways for future directions in safeguarding critical infrastructures and sensitive information. This section outlines the key challenges and provides recommendations for strengthening the kingdom's cyber resilience.

4.1. Building Cyber Capabilities

The surge in demand for skilled cybersecurity professionals is a global trend, and Saudi Arabia has recognized the urgent need to cultivate a robust cyber workforce. This initiative reflects a strategic commitment to not only address current security challenges but also to prepare for emerging cyber threats. Investing in education and training programs is critical for developing the necessary skills and expertise within the local population.

Educational Initiatives: Saudi Arabia has been implementing a range of educational programs, from university degrees in cybersecurity to specialized training courses and certifications [78], [93]. These programs are designed to equip students and professionals with up-to-date knowledge and practical skills in cybersecurity.

Public-Private Partnerships: The kingdom is also fostering partnerships between educational institutions and the private sector. These collaborations help tailor educational programs to meet the specific needs of the industry, ensuring that the training is relevant and that graduates are ready to enter the workforce.

Career Development Programs. Alongside formal education, Saudi Arabia is supporting career development initiatives such as internships, apprenticeships, and continuous professional development opportunities. These programs help bridge the gap between education and employment, providing hands-on experience and facilitating the transition into the cybersecurity field.

4.2. Evolving Threat Landscape

Saudi Arabia's cyber threat landscape is dynamic due to its strategic geopolitical position and economic reliance on oil and finance. The evolving nature of global cyber threats, with attackers refining methods and technologies, presents significant challenges for cybersecurity professionals and organizations in the region.

Sophistication of Attacks: Attackers are using more sophisticated methods, such as polymorphic malware that can change its code to evade detection, and advanced ransomware techniques that not only encrypt data but also steal it, threatening to release it publicly if the ransom is not paid.

Emergence of IoT Vulnerabilities: With the rapid expansion of the Internet of Things (IoT), many devices are connected to the internet without sufficient security measures. This expands the attack surface dramatically, giving cybercriminals new opportunities to exploit.

Increase in State-Sponsored Attacks: The region has seen a rise in state-sponsored cyber activities aimed at espionage, disruption of critical infrastructure, and influencing political and economic situations. These state-level attacks are particularly challenging due to the resources and advanced capabilities they possess.

Exploitation of Human Factors: Phishing attacks continue to evolve, targeting specific individuals (spear phishing) within organizations to gain sensitive information or access credentials. The sophistication of these attacks makes them harder to detect and prevent.

4.3. Regulatory and Compliance Issues

Navigating the complex landscape of cybersecurity regulations and ensuring compliance poses significant challenges for organizations in Saudi Arabia, especially those operating across multiple jurisdictions. This complexity is compounded by the rapid evolution of cyber threats, which often outpaces regulatory frameworks, requiring constant vigilance and adaptability from businesses.

Rapidly Changing Regulations: Cybersecurity regulations in Saudi Arabia are continually updated to address new cyber threats and technological advancements. Organizations must stay informed about these changes to remain compliant, which requires a dedicated compliance team or external consultants familiar with both local and international cyber laws.

Multi-Jurisdictional Compliance: For multinational corporations operating in Saudi Arabia, compliance becomes even more challenging. They must not only adhere to Saudi regulations but also manage compliance with international standards and regulations, such as the GDPR (General Data Protection Regulation) in Europe or the CCPA (California Consumer Privacy Act) in the United States. This necessitates a nuanced understanding of how these regulations intersect and impact their operations.

Industry-Specific Regulations: Certain sectors, such as finance and healthcare, are subject to more stringent cybersecurity regulations due to the sensitivity of the data they handle. Organizations in these sectors must implement specific security measures and compliance protocols, which can vary significantly from one industry to another.

Compliance Verification and Audits: Regular audits are necessary to ensure compliance with cybersecurity regulations. These audits can be resource-intensive and require organizations to provide extensive documentation and access to their security processes, which can be challenging to manage alongside day-to-day operations.

Cultural and Organizational Challenges: Embedding a culture of compliance within an organization is crucial yet challenging. It requires training employees on the importance of cybersecurity measures and ensuring they understand and follow the required procedures.

4.4. Insider Threats

Managing insider threats in Saudi Arabia presents a complex challenge due to the often subtle nature of such risks, which can stem from both malicious intent and accidental breaches. Insider threats are particularly concerning because they originate from within the organization, often involving individuals who have authorized access to sensitive information and critical systems. The insiders can be categorized into two groups:

- 1) **Malicious Insiders:** These are individuals within the organization who intentionally abuse their access to steal information, sabotage systems, or otherwise harm the organization. Their motivations can vary from financial gain to ideological beliefs or personal grievances.
- 2) **Accidental Insiders:** Accidental threats occur when employees inadvertently cause security breaches through negligence, lack of awareness, or errors. This might include mishandling sensitive data, falling prey to phishing attacks, or misconfiguring security settings.

The impact of insider threats can be devastating, leading to significant data breaches, financial losses, operational disruptions, and reputational damage. Insiders, having legitimate access, can act undetected for long periods, causing substantial harm. Detection is challenging as insider activities often mimic legitimate actions, and their psychological and behavioral complexities make them hard to predict. Organizational silos and concerns over employee trust and privacy further complicate management. Many organizations focus more on external threats and may lack comprehensive strategies or tools to address insider risks effectively.

4.5. Future Directions

The future of cybersecurity lies in the development and deployment of advanced predictive technologies that can detect and neutralize threats before they materialize. Machine learning and AI are expected to play significant roles in this area. More importantly, strengthening collaboration between the public and private sectors is crucial for sharing threat intelligence, best practices, and resources. This collaborative approach can lead to more resilient cybersecurity frameworks. Indeed, expanding cybersecurity education programs to cultivate a broader understanding of security practices across all levels of an organization is essential. This includes regular training and awareness programs to mitigate human-factor vulnerabilities.

There is a growing need for international cybersecurity norms to guide state and non-state behavior, reducing cyber conflicts. The focus is shifting from prevention to resilience, ensuring organizations can operate during and quickly recover from cyberattacks, which includes developing robust incident response and recovery plans.

By addressing these challenges, Saudi Arabia can enhance its cybersecurity infrastructure to better protect against sophisticated threats. This requires technological solutions and a strategic, coordinated approach involving policy, people, and processes.

5. Closely Related Work

The exploration of cybersecurity issues in Saudi Arabia has indeed garnered increasing interest from the academic and research community in recent years. This focus is driven by the nation's critical role in global energy markets and its rapid technological advancements, which make it a prominent target for cyberattacks.

Review Studies: [94] has contributed to the understanding of cyberattacks in Saudi Arabia through his brief overview of specific incidents. While Aljurayyed's work provides valuable insights into selected cyberattacks, it covers only six specific incidents, offering a snapshot rather than a comprehensive analysis. [95] presents a case study focuses on cyber-attacks in Saudi Arabia, specifically examining the impact of two prominent types of malware: Shamoon and Mamba Ransomware. These attacks have significantly targeted and disrupted the Saudi environment. Similarly, [24] presents a case study of attacks on Saudi Organization by malwares. The work concentrates on two particular malwares: Shamoon and Ransomware.

Cybersecurity Challenges: [6] presents an analysis of the cybersecurity landscape in Saudi Arabia, focusing on the significant challenges the nation has encountered. Their work underscores the importance of advancing cybersecurity protocols and strategies to protect against evolving threats in a region that has become a focal point for cyber activities. Considering the impact of COVID-19 on the cybersecurity landscape in Saudi Arabia, the work by [96] and [97] examine the impact of COVID-19 on the cybersecurity environment across various sectors, highlighting specific challenges and adaptations necessitated by the pandemic. In the context of smart cities, [98] investigates the obstacles and complexities involved in implementing cybersecurity measures within the smart cities of Saudi Arabia.

Cybersecurity Strategy: [99] provides a review that examines how Saudi Arabia's cybersecurity policy framework compares with those of the United States of America, India, and Singapore, highlighting both the similarities and differences. Following a more comprehensive approach, [100] analyzes Saudi Arabia's cybersecurity strategy in comparison with the National Cyber Security Strategies (NCSS) of fifteen other countries.

6. Conclusion

Over the past twelve years, Saudi Arabia has faced an increasingly sophisticated array of cyberattacks, reflecting broader global trends in cybersecurity threats. These incidents have targeted critical sectors, including government, energy, and finance, highlighting the strategic importance of cybersecurity as a pillar of national security. This study methodically records the major cyberattacks that have taken place in Saudi Arabia from 2012 through 2024.

Key Findings: The following key findings are concluded:

- The research underscores a significant evolution in the complexity and sophistication of cyber threats in Saudi Arabia. Cyberattacks have progressively incorporated advanced technologies and strategic approaches to exploit vulnerabilities at both technical and human levels.
- The analysis reveals that these cyberattacks have not only threatened national security but also caused substantial economic disruptions. This underscores the interconnected nature of cybersecurity with the overall stability and prosperity of the nation.

- There has been a robust response from both the Saudi government and the private sector, involving the strengthening of cybersecurity policies, the implementation of advanced technological solutions, and significant investments in human capital. These efforts are pivotal in building a resilient cybersecurity framework.

Recommendations: Based on the findings from this longitudinal analysis, several recommendations emerge:

- Saudi Arabia should continue to invest in cutting-edge cybersecurity technologies and frameworks to stay ahead of potential threats. This includes adopting artificial intelligence and machine learning for predictive threat analysis.
- Strengthening the human aspect of cybersecurity through ongoing education and awareness campaigns is crucial. These programs should target all levels of organizations and the general public to cultivate a strong culture of cybersecurity.
- Given the global nature of cyber threats, enhancing international cooperation is vital. Participating in global cybersecurity initiatives can help Saudi Arabia both contribute to and benefit from shared knowledge, resources, and strategies.
- The study also highlights the need for robust regulatory frameworks that can adapt to the dynamic cybersecurity landscape. These should facilitate not only compliance but also encourage proactive security practices.

While Saudi Arabia has made significant strides in combating cyber threats, the journey towards cybersecurity is ongoing. In 2017, the country ranked 46th globally in cybersecurity according to the International Telecommunication Union (ITU) Global Cybersecurity Index (GCI) [101] and also ranked 13th overall in 2018 [102]. However, by 2020, Saudi Arabia reached the 2nd position in the world, ranking number 1 in the Arab States region [103]. This study not only contributes to a deeper understanding of the cybersecurity challenges faced during this journey but also provides a blueprint for future efforts to enhance national and organizational cybersecurity postures.

References:

- [1]. Saudi Vision 2030. (2024). *Saudi Vision 2030*. Vision2030. Retrieved from: <https://www.vision2030.gov.sa/en/> [accessed: 03 July 2024]
- [2]. Obaid, R. (2021). *Saudi Arabia had 7 million cyberattacks in 2021*. Arab News. Retrieved from: <https://www.arabnews.com/node/1832526/saudi-arabia> [accessed: 03 July 2024]
- [3]. Council on Foreign Relations. (2012). *Compromise of Saudi Aramco and RasGas*. Crf.org. Retrieved from: <https://www.cfr.org/cyber-operations/compromise-saudi-aramco-and-rasgas> [accessed: 04 August 2024]
- [4]. Arabian Business. (2024). *160,000 cyberattacks a day in Saudi Arabia*. Arabian Business. Retrieved from: <https://www.arabianbusiness.com/160-000-cyberattacks-day-in-saudi-arabia-630120.html> [accessed: 05 July 2024]
- [5]. Kelly, C. (2020). *IBM: Cyber breaches cost enterprises in the UAE and KSA over \$6.5M per attack in 2020*. ITP News. Retrieved from: <https://www.itp.net/news/93473-ibm-cyber-breaches-cost-enterprises-in-the-uae-and-ksa-over-65m-per-attack-in-2020> [accessed: 06 July 2024]
- [6]. Quadri, A., & Khan, M. K. (2019). *Cybersecurity challenges of the Kingdom of Saudi Arabia*. Global Foundation for Cyber Studies and Research.
- [7]. The Daily Swig. (2023). *Iranian cyber-threat groups make up for lack of technical sophistication with social engineering trickery*. Portswigger.net. Retrieved from: <https://portswigger.net/daily-swig/iranian-cyber-threat-groups-make-up-for-lack-of-technical-sophistication-with-social-engineering-trickery> [accessed: 07 July 2024]
- [8]. SOCRadar Cyber Intelligence Inc. (2023). *Saudi Arabia (KSA) threat landscape report*. Socradar.io. Retrieved from: <https://socradar.io/resources/country-reports/saudi-arabia-ksa-threat-landscape-report/> [accessed: 08 July 2024]
- [9]. Arab News. (2023). *Saudi, UAE organizations prime targets of cybercrime in GCC*. Arab News. Retrieved from: <https://www.arabnews.com/node/2235636/media> [accessed: 09 July 2024]
- [10]. Mokshita, P. (2024). *MENA's agile talent pool ranks highest in expertise*. SME10X. Retrieved from: <https://www.sme10x.com/10x-industry/menas-agile-talent-pool-ranks-highest-in-expertise> [accessed: 10 July 2024]
- [11]. Buller, A. (2020). *Saudi Arabia sees cyber security boom as coronavirus bites*. Computer Weekly. Retrieved from: <https://www.computerweekly.com/news/252489175/Saudi-Arabia-sees-cyber-security-boom-as-coronavirus-bites> [accessed: 14 July 2024]
- [12]. National Cybersecurity Authority (NCA). (2024). *National Cybersecurity Authority of Saudi Arabia*. Nca.gov.sa. Retrieved from: <https://nca.gov.sa/en> [accessed: 15 July 2024]

- [13]. Saudi National Cybersecurity Authority. (2024). *Computer Emergency Response Team (CERT)*. Retrieved from: <https://cert.gov.sa/en> [accessed: 17 July 2024]
- [14]. Dark Reading. (2023). *Saudi Arabia cyber capabilities ranked second globally*. Dark Reading. Retrieved from: <https://www.darkreading.com/cyber-risk/saudi-arabia-cyber-capabilities-ranked-second-globally> [accessed: 19 July 2024]
- [15]. Kitchenham, B., & Charters, S. (2007). *Guidelines for performing systematic literature reviews in software engineering*. Keele University & Durham University.
- [16]. Höst, M., & Oručević-Alagić, A. (2011). A systematic review of research on open source software in commercial software product development. *Information and Software Technology*, 53(6), 616-624.
- [17]. Jalali, S., & Wohlin, C. (2010). Agile practices in global software engineering-A systematic map. In *2010 5th IEEE International Conference on Global Software Engineering*, 45-54. IEEE.
- [18]. Rao, V. M., & Singh, Y. P. (2013). Decision Tree Induction for Financial Fraud Detection. *Proceeding of the International conference on artificial intelligence in computer science and ICT (AICS 2013)*, 321-328.
- [19]. Bronk, C., & Tikk-Ringas, E. (2013). The cyber attack on Saudi Aramco. *Survival*, 55(2), 81-96.
- [20]. Zhioua, S. (2013). The middle east under malware attack dissecting cyber weapons. In *2013 IEEE 33rd International Conference on Distributed Computing Systems Workshops*, 11-16.
- [21]. GReAT. (2014). *The Regin platform: Nation-state ownage of GSM networks*. Kaspersky Lab.
- [22]. Symantec. (2014). *Top-tier espionage tool enables stealthy surveillance* (White paper). Symantec.
- [23]. Symantec, D. (2014). *Cyberespionage attacks against energy suppliers, version 1.21*. Mountain View, California.
- [24]. Alelyani, S., & GR, H. K. (2018). Overview of cyberattack on saudi organizations. *Journal of Information Security and Cybercrimes Research*, 1(1), 32-39.
- [25]. Malware News. (2017). *The State of Shamoon: Same Actor, Different lines*. Malware news. Retrieved from: <https://malware.news/t/the-state-of-shamoon-same-actor-different-lines/11146> [accessed 25 July 2024].
- [26]. The New York Times. (2018). *A Cyberattack in Saudi Arabia Had a Deadly Goal. Experts Fear Another Try*. The New York Times. Retrieved from: <https://www.nytimes.com/2018/03/15/technology/saudi-arabia-hacks-cyberattacks.html> [accessed: 25 July 2024].
- [27]. Mascellino, A. (2024). *Sophisticated cyber-attack hits Islamic charity in Saudi Arabia*. Infosecurity Magazine. Retrieved from: <https://www.infosecurity-magazine.com/news/cyberattack-hits-islamic-charity/> [accessed: 26 July 2024].
- [28]. Malley, R. (2019). The unwanted wars. *Foreign Affairs*, 98, 38.
- [29]. Greenley, H. L., Blanchard, C. M., & Ratner, M. (2019). *Attacks on Saudi oil facilities: Effects and responses*. Congressional Research Service (CRS).
- [30]. Kaspersky. (2019). *2019 Cyber Defense Summit*. Kaspersky. Retrieved from: <https://me-en.kaspersky.com/about/press-releases/2019cyber-defense-summit> [accessed: 27 July 2024]
- [31]. IBM. (2024). *Cost of a Data Breach Report 2024*. IBM. Retrieved from: <https://www.ibm.com/security/digital-assets/cost-data-breach-report/#/> [accessed: 18 August 2024].
- [32]. VMware. (2021). *2021 Global Security Insight Report*. VMware. Retrieved from: <https://www.vmware.com/docs/vmwcb-infographic-global-security-insights-report-2021> [accessed: 29 July 2024]
- [33]. Positive Technologies. (2023). *Cybersecurity threatscape in the Middle East: 2022–2023*. Positive Technologies. Retrieved from: <https://global.ptsecurity.com/analytics/cybersecurity-threatscape-in-the-middle-east-2022-2023> [accessed: 02 August 2024]
- [34]. Dark Reading. (2022). *Undetected attacks against Middle East conducted*. Dark Reading. Retrieved from: <https://www.darkreading.com/cyberattacks-data-breaches/undetected-attacks-against-middle-east-conducted> [accessed: 02 August 2024]
- [35]. Kaspersky ICS CERT. (2022). *Threat landscape for industrial automation systems: Statistics for H2 2022*. Retrieved from: <https://ics-cert.kaspersky.com/publications/reports/2023/03/06/threat-landscape-for-industrial-automation-systems-statistics-for-h2-2022/> [accessed: 04 August 2024]
- [36]. GROUP-IB. (2022). *Hi-tech crime trends 2022-2023*. GROUP-IB. Retrieved from: <https://www.group-ib.com/resources/research-hub/hi-tech-crime-trends-2022/> [accessed: 05 August 2024]
- [37]. Sirar by STC. (2023). *Threat landscape report in 2023*. Retrieved from: <https://www.sirar.com.sa/wp-content/uploads/2024/02/Threat-Landscape-Report-in-2023-Final.pdf> [accessed: 05 August 2024]
- [38]. Infosecurity Magazine. (2023). *Cyberattack hits Islamic charity*. Infosecurity Magazine. Retrieved from: <https://www.infosecurity-magazine.com/news/cyberattack-hits-islamic-charity/> [accessed: 06 August 2024].
- [39]. Positive Technologies. (2024). *How APT groups operate in the Middle East*. Positive Technologies. Retrieved from: <https://global.ptsecurity.com/analytics/how-apt-groups-operate-in-the-middle-east> [accessed: 06 August 2024]
- [40]. Symantec Threat Hunter Team. (2016, 30. november). *Shamoon: Back from the dead and destructive as ever*. Symantec Enterprise Blogs. Retrieved from: <https://symantec-enterprise-blogs.security.com/blogs/threat-intelligence/shamoon-back-destructive> [accessed: 07 August 2024]
- [41]. Tarakanov, D. (2012). *Shamoon the Wiper in details*. Securelist by Kaspersky. Retrieved from: <https://securelist.com/shamoon-the-wiper-in-details/34369/> [accessed: 08 August 2024]
- [42]. Dehlawi, Z., & Abokhodair, N. (2013). Saudi Arabia's response to cyber conflict: A case study of the Shamoon malware incident. *2013 IEEE International Conference on Intelligence and Security Informatics*, 73-75.

- [43]. Reuters. (2017). *Saudi Arabia warns on cyber defense as Shamoon resurfaces*. Reuters. Retrieved from: <https://www.reuters.com/article/us-saudi-cyber/saudi-arabia-warns-on-cyber-defense-as-shamoon-resurfaces-idUSKBN1571ZR/> [accessed: 09 August 2025]
- [44]. Groll, E. (2017). *Cyber attack targets safety system at Saudi Aramco*. Foreign Policy. Retrieved from: <https://foreignpolicy.com/2017/12/21/cyber-attack-targets-safety-system-at-saudi-aramco/> [accessed: 09 August 2024]
- [45]. Center for Strategic and International Studies (CSIS). (2014). *Net losses: Estimating the global cost of cybercrime*. CSIS. Retrieved from: <https://www.csis.org/analysis/net-losses-estimating-global-cost-cybercrime> [accessed: 10 August 2024]
- [46]. IBM. (2022). *What is endpoint detection and response (EDR)?*. IBM. Retrieved from: <https://www.ibm.com/topics/edr> [accessed: 10 August 2024]
- [47]. Alabdulatif, A. (2018). *Cybercrime and analysis of laws in the Kingdom of Saudi Arabia*. [Doctoral dissertation University of Houston].
- [48]. Kaspersky. (2017). *Double trouble: A pair of wipers in Saudi Arabia*. Kaspersky. Retrieved from: <https://www.kaspersky.com/blog/shamoon-stonedrill/15170/> [accessed: 02 July 2024].
- [49]. Albano, K., & Kessem, L. (2017). *The full Shamoon: How the devastating malware was inserted into networks*. IBM. Retrieved from: <https://securityintelligence.com/the-full-shamoon-how-the-devastating-malware-was-inserted-into-networks/> [accessed: 03 July 2024].
- [50]. Derbyshire, R., et al. (2018). An analysis of cyber security attack taxonomies. *2018 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)*, 153-161.
- [51]. National Cybersecurity Authority (NCA). (2024). *About the National Cybersecurity Authority (NCA)*. NCA. Retrieved from: <https://nca.gov.sa/en/> [accessed: 04 August 2024]
- [52]. National Institute of Standards and Technology (NIST). (2024). *Glossary term: RFC (Request for Comments)*. NIST Computer Security Resource Center. Retrieved from: <https://csrc.nist.gov/glossary/term/rfc> [accessed: 05 August 2024]
- [53]. International Organization for Standardization (ISO). (2022). *ISO/IEC 27001:2022: Information security, cybersecurity, and privacy protection – Information security management systems – Requirements*. ISO. Retrieved from: <https://www.iso.org/standard/27001> [accessed: 06 August 2024]
- [54]. Internet Assigned Numbers Authority (IANA). (2024). *Home page*. Iana. Retrieved from: <https://www.iana.org/> [accessed: 06 August 2024]
- [55]. Tanchum, M. E. (2019). Drone attacks on Saudi Oil Infrastructure are a calibrated message from Iran. *IAI Commentaries*, 19, 35.
- [56]. Ali, I. & Stewart, P. (2019). *Exclusive: U.S. carried out secret cyber strike on Iran in wake of Saudi oil attack – officials*. Reuters.
- [57]. Said, S., Malsin, J., & Donati, J. (2019). U.S. Blames Iran for Attack on Saudi Oil Facilities. *The Wall Street Journal*.
- [58]. Kearney, L. (2019). *Oil jumps nearly 15% in record trading after attack on Saudi facilities*. Reuters. Retrieved from: <https://www.reuters.com/article/us-global-oil/oil-prices-surge-15-after-attack-on-saudi-facilities-hits-global-supply-idUSKBN1W00UG/> [accessed: 08 August 2024]
- [59]. National Cybersecurity Authority (NCA). (2024). *The National Cybersecurity Strategy (Overview)*. NCA.
- [60]. Tactical Report. (2023). *Saudi Arabia expanded cybersecurity cooperation with the U.S*. Tactical Report. Retrieved from: <https://www.tacticalreport.com/daily/62342-saudi-arabia-expanded-cybersecurity-cooperation-with-the-us> [accessed: 09 August 2024]
- [61]. Monaghan, S. (2019). Countering hybrid warfare. *Prism*, 8(2), 82-99.
- [62]. Gonçalves, C. P. (2019). *Cyberspace and Artificial Intelligence: The New Face of Cyber-Enhanced Hybrid Threats*. *Cyberspace*.
- [63]. Bachmann, S. D. (2011). Hybrid threats, cyber warfare and NATO's comprehensive approach for countering 21st century threats-mapping the new frontier of global risk and security management. *Amicus Curiae*, 88, 24.
- [64]. Aaronson, M., et al. (2011). NATO countering the hybrid threat. *Prism*, 2(4), 111-124.
- [65]. Elonheimo, T. (2021). Comprehensive security approach in response to Russian hybrid warfare. *Strategic Studies Quarterly*, 15(3), 113-137.
- [66]. Olech, A., & Siekierka, K. (2021). *Cybersecurity in Saudi Arabia*. Institute of New Europe (INE).
- [67]. National Cybersecurity Authority. (2019). *Critical Systems Cybersecurity Controls (CSCC – 1: 2019)*. NCA. Retrieved from: https://cdn.nca.gov.sa/api/files/public/upload/f15af01c-de59-4281-95e2-03a770655937_Critical-Systems-Cybersecurity-Controls.pdf [accessed: 10 August 2024]
- [68]. National Cybersecurity Authority. (NCA). (n.d.). *Cybersecurity toolkits*. NCA. Retrieved from: <https://nca.gov.sa/en/legislation?item=665&slug=guide-lines-list> [accessed: 10 August 2024]
- [69]. Fast Digital Technology. (2024). *EDR solution in KSA*. Fast Digital Technology. Retrieved from: <https://fastdigitaltechnology.com/cyber-threat/edr-solution> [accessed: 21 July 2024]
- [70]. Infratech. (2024). *Endpoint security*. Infratech. Retrieved from: <https://www.infratech.com.sa/endpoint-security/> [accessed: 22 July 2024]
- [71]. National Cybersecurity Authority. (2024). *Regulation Documents*. NCA.
- [72]. NordVPN. (2024). *NordVPN survey reveals: Users still trust free VPNs*. NordVPN. Retrieved from: <https://atlasvpn.com/vpn-adoption-index> [accessed: 25 July 2024]
- [73]. Statista. (2020, februar). *Public cloud services market value in the Middle East and North Africa (MENA region) from 2015 to 2021, by cloud segment (in million U.S. dollars)*. Statista. Retrieved from: <https://www.statista.com/statistics/575559/mena-public-cloud-services-market-revenue-by-segment/> [accessed: 26 July 2024]

- [74]. Sharma, A. (2024). *Hacker posts exploits for over 49,000 vulnerable Fortinet VPNs*. Bleeping Computer. Retrieved from: <https://www.bleepingcomputer.com/news/security/hacker-posts-exploits-for-over-49-000-vulnerable-fortinet-vpns/#google%20vignette> [accessed: 27 July 2024]
- [75]. Cybersecurity and Infrastructure Security Agency (CISA). (2022). *Cybersecurity Advisory - 2021 Top Routinely Exploited Vulnerabilities*. Retrieved from: <https://www.cisa.gov/news-events/cybersecurity-advisories/aa22-117a> [accessed: 28 July 2024]
- [76]. IBM Security. (2021). *X-Force Threat Intelligence Index*. IBM.
- [77]. Verizon business. (2023). *2023 Data Breach Investigations Report*. Verizon. Retrieved from: <https://www.verizon.com/business/resources/reports/dbir/> [accessed: 29 July 2024]
- [78]. National Cybersecurity Authority (NCA). (2019). *Critical systems cybersecurity controls (CSCCS)*. NCA. Retrieved from: <https://www.nca.gov.sa/en/legislation?item=194&slug=controls-list> [accessed: 29 July 2024]
- [79]. Taqnia Institute. (2024). *Developing Workforces Through Knowledge*. Taqniacyber. Retrieved from: <https://www.taqniacyber.com/TaqniaInstitute.html> [accessed: 30 July 2024]
- [80]. National Cybersecurity Authority (NCA). (2022). *The National Cybersecurity Authority (NCA) launches the 'Cyberic' program for the development of the cybersecurity sector in the Kingdom of Saudi Arabia*. NCA. Retrieved from: <https://nca.gov.sa/en/news?item=234> [accessed: 03 August 2024]
- [81]. National Cybersecurity Authority (NCA). (2023). *Essential cybersecurity controls (ECCS)*. NCA. Retrieved from: <https://nca.gov.sa/en/legislation?item=191&slug=controls-list> [accessed: 03 August 2024]
- [82]. National Cybersecurity Authority (NCA). (2022). *Operational technology cybersecurity controls (OTCCS)*. NCA. Retrieved from: <https://www.nca.gov.sa/en/legislation?item=195&slug=controls-list> [accessed: 04 August 2024]
- [83]. Statista. (2024). *Percentage of employees who work from home all or most of the time worldwide from 2015 to 2023 [graph]*. Statista. Retrieved from: <https://www.statista.com/statistics/1450450/employees-remote-work-share/> [accessed: 05 August 2024]
- [84]. IBM. (n.d.). *Cost of a data breach report 2023*. IBM. Retrieved from: <https://www.ibm.com/reports/data-breach> [accessed: 05 August 2024]
- [85]. Saudi Press Agency. (2022). *The National Cybersecurity Authority organizes an awareness session for employees of the Ministry of Information*. Spa.gov. Retrieved from: <https://www.spa.gov.sa/2390941> [accessed: 07 August 2024]
- [86]. Okaz. (2022). *The National Cybersecurity Authority launches the national cybersecurity awareness campaign*. Okaz. Retrieved from: <https://www.okaz.com.sa/news/local/2115858> [accessed: 08 August 2024]
- [87]. CrowdStrike. (2023). *CrowdStrike 2024 global threat report*. CrowdStrike. Retrieved from: <https://go.crowdstrike.com/global-threat-report-2024-thank-you.html> [accessed: 09 August 2024]
- [88]. Lemos, R. (2024). *Saudi Arabia, UAE top list of APT-targeted nations in the Middle East*. Dark Reading. Retrieved from: <https://www.darkreading.com/vulnerabilities-threats/saudi-arabia-uae-top-list-of-apt-targeted-nations-in-middle-east> [accessed: 30 July 2024]
- [89]. Lemos, R. (2024). *UAE, Saudi Arabia become plum cyberattack targets*. Dark Reading. Retrieved from: <https://www.darkreading.com/cyberattacks-data-breaches/uae-saudi-arabia-cyberattack-targets> [accessed: 01 August 2024]
- [90]. MITRE. (2024). *Active scanning*. Attack Mitre. Retrieved from: <https://attack.mitre.org/techniques/T1595/> [accessed: 02 August 2024]
- [91]. MITRE. (2024). *Gather victim network information: IP addresses*. Attack Mitre. Retrieved from: <https://attack.mitre.org/techniques/T1590/005/> [accessed: 03 August 2024]
- [92]. MITRE. (2024). *Gather victim org information: Determine physical locations*. Attack Mitre. Retrieved from: <https://attack.mitre.org/techniques/T1591/001/> [accessed: 03 August 2024]
- [93]. National Cybersecurity Authority (NCA). (2023). *Alignment with the Saudi framework for higher education in cybersecurity (Cyber-Education)*. NCA. Retrieved from: <https://nca.gov.sa/pages?item=92d19870-8e6f-4d51-aaff-3890763e8f7e> [accessed: 05 August 2024]
- [94]. Aljuryyed, A. (2022). *Cybersecurity Issues in the Middle East: Case Study of the Kingdom of Saudi Arabia. Cybersecurity Capabilities in Developing Nations and Its Impact on Global Security*, 62-82. IGI Global Scientific Publishing.
- [95]. Al-Mulhim, R. A., et al. (2020). *Cyber-attacks on Saudi Arabia environment*. *International Journal of Computer Networks and Communications Security*, 8(3), 26-31.
- [96]. Alotibi, G. N., & Al Abdulwahid, A. (2023). *An investigation of cybersecurity issues of remote work during the COVID-19 pandemic in Saudi Arabia*. *International Journal of Advanced Computer Science and Applications*, 14(1).
- [97]. Almrezeq, N., et al. (2023). *Cyber security attacks and challenges in Saudi Arabia during COVID-19*. *International Journal of Computer Science & Network Security*, 23(10), 179-187.
- [98]. Alhalafi, N., & Veeraraghavan, P. (2023). *Exploring the challenges and issues in adopting cybersecurity in Saudi smart cities: Conceptualization of the cybersecurity-based UTAUT model*. *Smart Cities*, 6(3), 1523-1544.
- [99]. Alhalafi, N., & Veeraraghavan, P. (2021). *Cybersecurity policy framework in Saudi Arabia: Literature review*. *Frontiers in Computer Science*, 3, 736874.
- [100]. Alghamdi, A. A. (2022). *A Comparative Analysis to Advancing the National Cybersecurity Strategy in Saudi Arabia*. *Journal of Engineering and Applied Sciences*, 9(1).
- [101]. International Telecommunication Union (ITU). (2017). *Global cybersecurity index (GCI)*. ITU.
- [102]. International Telecommunication Union (ITU). (2018). *Global cybersecurity index (GCI)*. ITU.
- [103]. International Telecommunication Union (ITU). (2020). *Global cybersecurity index (GCI)*. ITU.